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MODULE 07

Kubernetes and OpenShift

Day 2 · Core MVP Module

ADOBE STOCK PLACEHOLDER

Suggested search: “cloud infrastructure network diagram (wide hero banner)”

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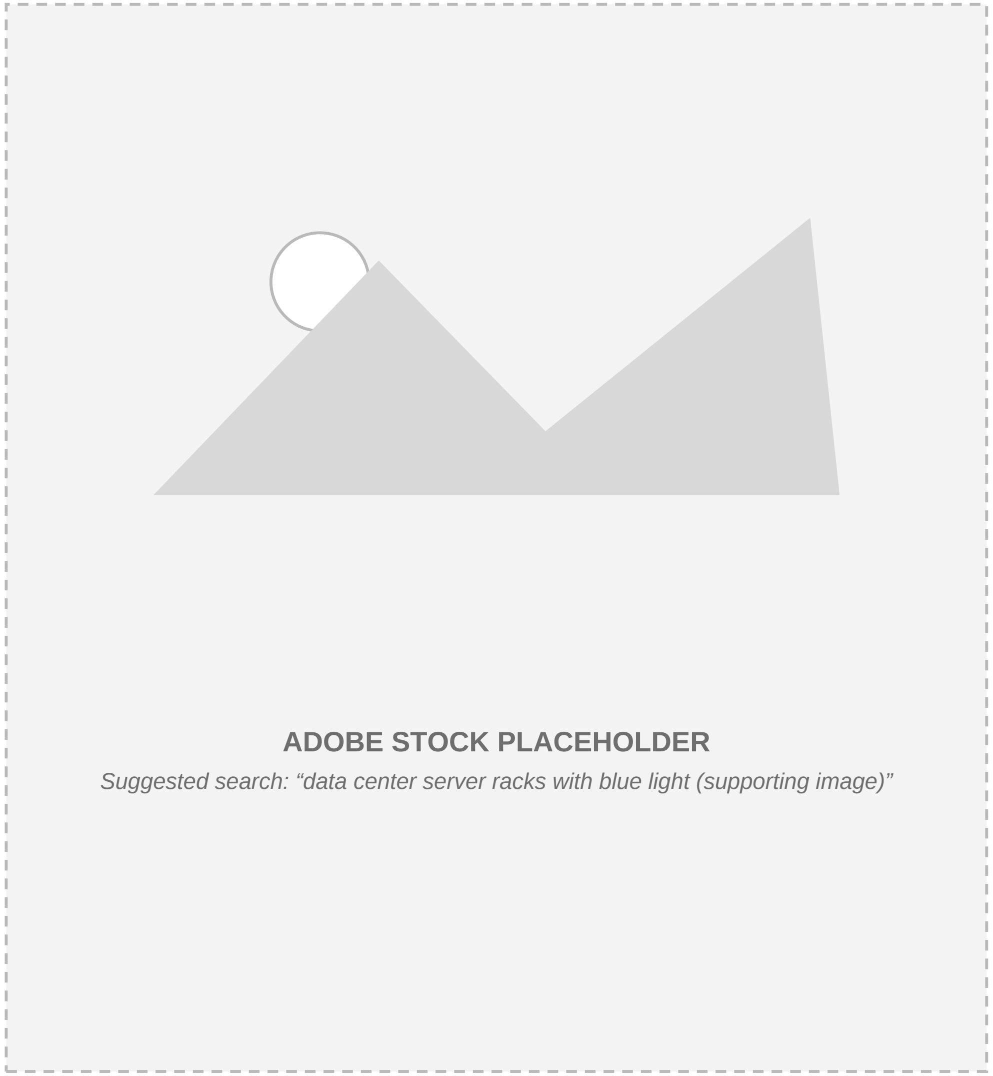
LEARNING OBJECTIVES

- Deploy workloads to Kubernetes or OpenShift.
- Configure health probes, configuration objects, and scaling policy.
- Verify deployment health through route or ingress checks.



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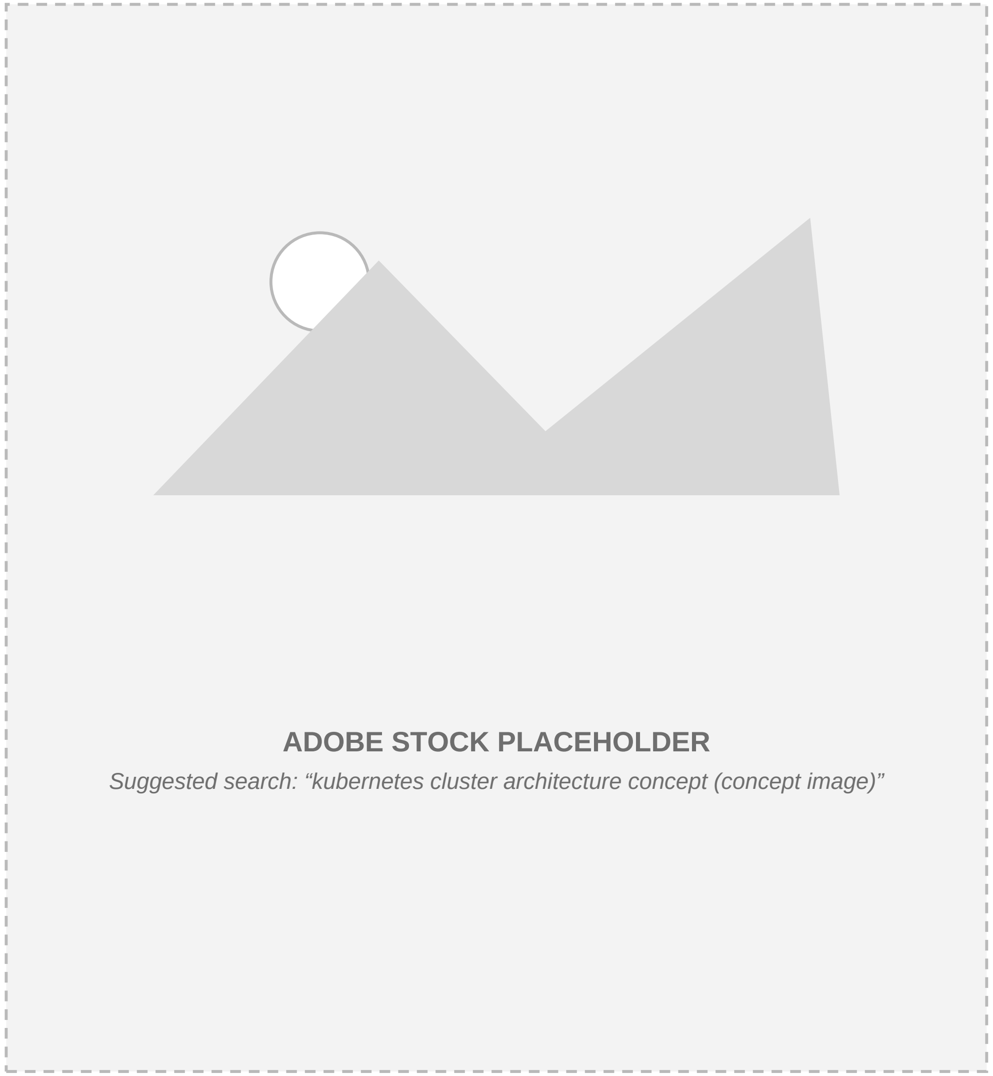
WHY THIS MODULE MATTERS

This module operationalizes the container image by deploying it into a managed cluster context.

Learners validate reliability behavior once the workload is running.



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CONCEPT MAP

What actually gates traffic to a newly deployed pod, per the Kubernetes docs.

1

DEPLOY POD

Container scheduled onto the cluster

→

2

READINESS PROBE

Gates traffic until the app can serve it

→

3

LIVENESS PROBE

Restarts the container if it hangs

→

4

ROUTE / INGRESS

Exposes the healthy pod externally

→

5

AUTOSCALE

HPA adds/removes pods under load

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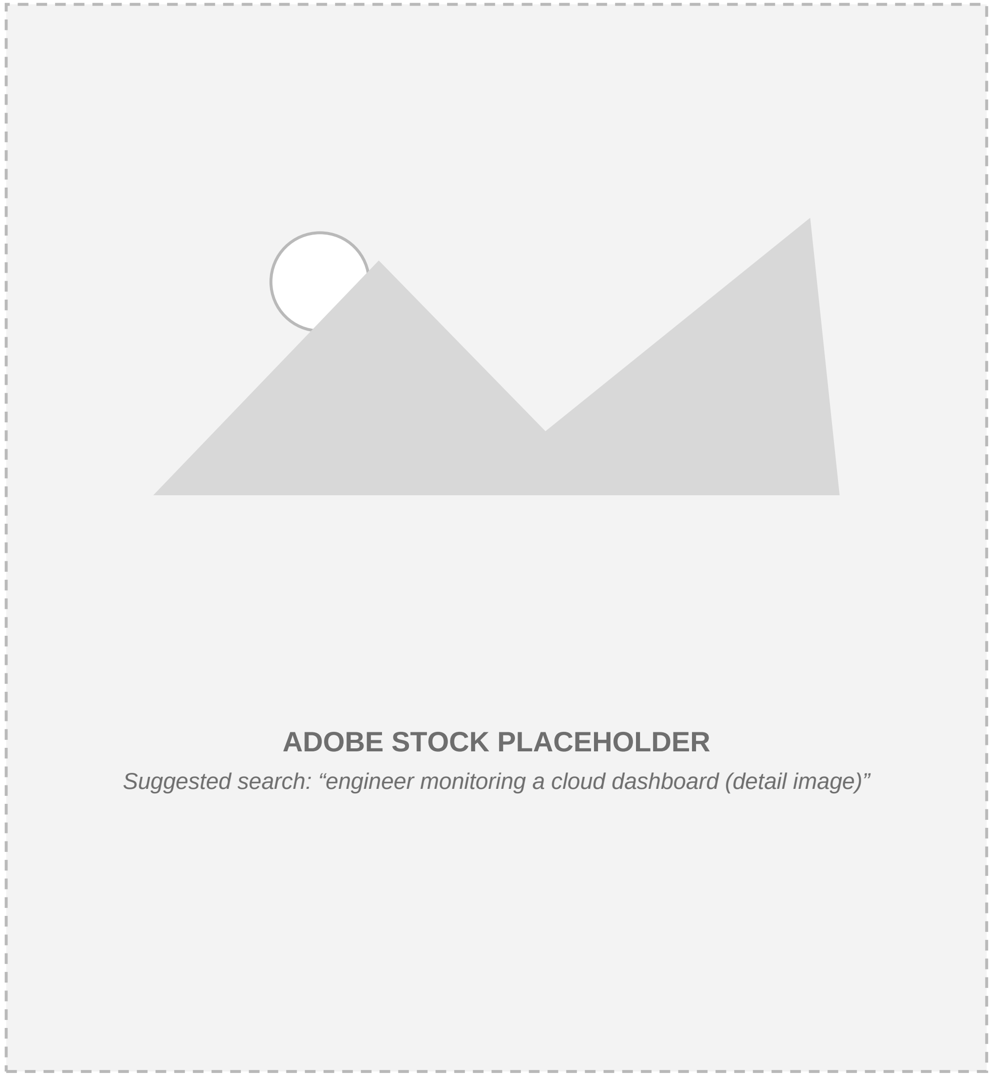
FROM THE DOCS: PROBES, PRECISELY

- Liveness probe: checks whether a container is still running correctly (e.g., not deadlocked).
- Liveness failure: once failures cross the threshold, the kubelet restarts the container.
- Readiness probe: checks whether a container is ready to accept traffic.
- Readiness failure: the Pod's IP is pulled from Service endpoints — traffic stops, container keeps running.
- OpenShift Routes add features plain Kubernetes Ingress lacks, e.g. TLS re-encryption and blue-green traffic splits.



Source: Liveness, Readiness, and Startup Probes — Kubernetes — https://kubernetes.io/docs/concepts/workloads/pods/probes/

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COURSEWARE FOCUS

- Kubernetes primitives
- OpenShift specifics
- Routes
- Probes
- Scaling

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REAL DEPLOY COMMANDS — S2I-DOTNETCORE-EX

REAL CODE — README.adoc (Red Hat's official S2I .NET sample)

```
$ oc new-project mydemo

$ oc apply -f https://raw.githubusercontent.com/redhat-developer/
    s2i-dotnetcore/main/dotnet_imagestreams.json

$ oc new-app dotnet:10.0~https://github.com/redhat-developer/
    s2i-dotnetcore-ex#dotnet-10.0 --context-dir app

$ oc expose service s2i-dotnetcore-ex
$ oc get route s2i-dotnetcore-ex
```

The exact source-to-image build-and-deploy flow Lab 07 follows against OpenShift.

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TOOLS & RESOURCES

Workshop Repo

Azure-Samples/aro-eshop-workshop (Azure Red Hat OpenShift Workshop)

Manifest files (.yaml), OpenShift Routes, DeploymentConfigs, and ImageStreams.

Sandbox Environment

Red Hat Developer Sandbox for OpenShift

Free 30-day cluster environment for students to deploy containerized workloads.

K8s Guides

Kubernetes Documentation: ConfigMaps, Secrets, Probes, Autoscaling

Configuring liveness/readiness probes, secrets, and autoscaling.

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SUPPORTING ASSETS

- OpenShift sample: course/repos/s2i-dotnetcore-ex
- Container image from Module 6

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Suggested search: “data center server racks with blue light (reference screenshot)”

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LAB 07: KUBERNETES AND OPENSHIFT

ADOBE STOCK PLACEHOLDER

Suggested search: “kubernetes cluster architecture concept (lab / hands-on photo)”

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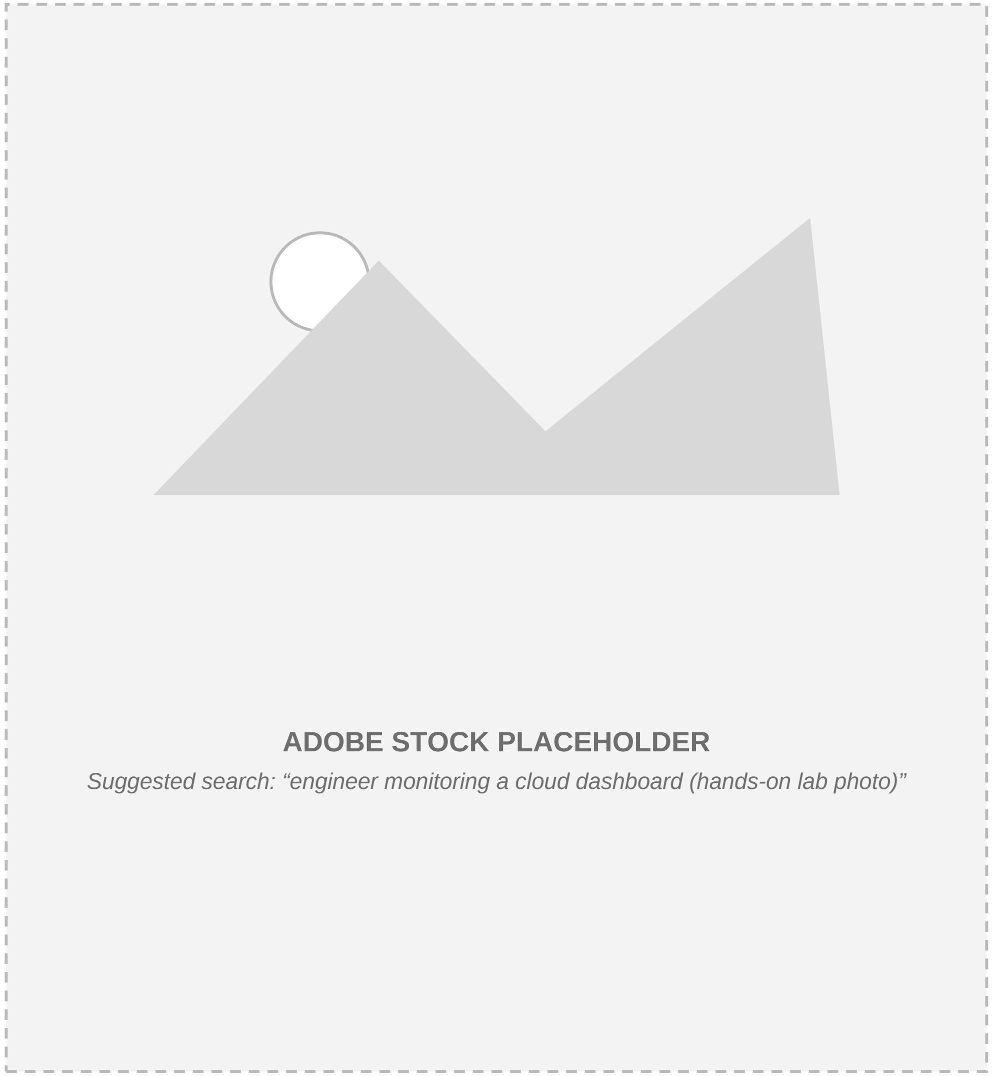
LAB STEPS

Goal: Deploy the containerized app to cluster infrastructure and validate reliability.

Tools: kubectl or oc · Cluster: OpenShift, or AKS as a fallback.

- Step 1 — Create a namespace and apply baseline manifests.
- Step 2 — Configure ConfigMaps and Secrets.
- Step 3 — Add readiness and liveness probes.
- Step 4 — Expose the service via route or ingress.
- Step 5 — Configure and test autoscaling.
- Validation: pod/deployment health is green, the route is reachable, and autoscaling responds to load.



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KNOWLEDGE CHECK: Kubernetes and OpenShift

What is the purpose of a Kubernetes readiness probe?

A. To restart the container on a fixed schedule

B. To tell Kubernetes when a container is ready to accept traffic

C. To scan the container image for vulnerabilities

D. To assign the pod a static IP address

ADOBE STOCK PLACEHOLDER

Suggested search: “cloud infrastructure network diagram (classroom discussion photo)”

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KNOWLEDGE CHECK: Kubernetes and OpenShift

What happens specifically when a Kubernetes readiness probe fails?

A. The container is immediately deleted

B. The pod's IP is removed from Service endpoints, but the container keeps running

C. The whole node is cordoned

D. Nothing happens

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SOURCES & FURTHER READING

EXTERNAL DOCUMENTATION

- **Liveness, Readiness, and Startup Probes — Kubernetes**
  https://kubernetes.io/docs/concepts/workloads/pods/probes/

REFERENCED IN THIS REPOSITORY

- `course/repos/s2i-dotnetcore-ex/README.adoc`
- `course/mvp-delivery/labs/lab-07-kubernetes-openshift.md`

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RECAP & SUCCESS CRITERIA

- Workload is reachable and healthy, and can scale under load.
- Tier: Core MVP Module

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